IDG Sovereign Architecture Blueprint & Standards

Comprehensive Technical Specification for In-Country Sovereign Cloud Hosting and Resilient Enterprise Zero-Trust Infrastructure

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1. Executive Mandate & National Objectives

Iraq Digital Gateway (IDG) establishes sovereign technical independence across critical institutional infrastructure. This blueprint mandates that all institutional records, customs manifests, biometric identity tokens, and financial settlements must reside strictly within in-country sovereign data centers within Iraq’s geographical borders.

2. Tri-Regional Geographic Topology (Baghdad, Erbil, Basra)

To achieve high availability (99.999% uptime) and geographic disaster tolerance, IDG operates an active-active-passive mesh topology across three strategic nodes:

â¢ Baghdad Central Node (BG-DC1): Primary institutional transaction routing and CBI RTGS gateway.

â¢ Erbil Northern Node (EBL-DC2): Sovereign analytics, ZaNa Academy computing, and active backup.

â¢ Basra Maritime Node (BSR-DC3): Port manifest edge caching and Umm Qasr NSW edge controller.

Inter-node synchronization utilizes sovereign dark fiber with dual-layer AES-256-GCM encryption.

3. Zero-Trust Boundary & HSM Hardware Key Custody

Perimeter-only defense is explicitly rejected. IDG enforces strict Zero-Trust Architecture (ZTA):

â¢ Mutual TLS (mTLS 1.3) required for every intra-service microservice transaction.

â¢ FIPS 140-3 Level 4 Hardware Security Modules (HSM) maintain root cryptographic keys.

â¢ Key custody resides strictly under multi-party institutional threshold authorization.

â¢ Foreign cloud providers are barred from holding or decrypting sovereign state records.

4. Air-Gapped Core Protocols & Anti-Vendor Lock-in

Foundation registries (Citizen Identity, Land Corridors, National Single Window Ledger) are isolated behind unidirectional data diodes (air-gapped replication channels). All interfaces adhere strictly to open standards (OAS 3.1, gRPC, W3C DID, ISO 20022). Proprietary vendor APIs that induce vendor lock-in are barred from sovereign core systems.

5. Autonomous National Survivability

In the event of an international submarine fiber cut or geopolitical network embargo, the national sovereign cloud continues full autonomous local operation without loss of customs clearance, financial settlement, or institutional identity verification.

6. Architectural Governance Sign-off & Verification

This specification has been reviewed and certified by the IDG Technical Taskforce. Periodic compliance audits occur every 90 days in coordination with National Security.

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[VERIFIED] Cryptographic Manifest Hash: 9f8a4e21b7c0d381ea6249b5c2198f7e